RFID Based Cycle Parking Stand With Authenticated Locking System

ALDRIN G

May 27, 2023

Summer Project 2023

Project Details

- TITLE : RFID Based Cycle Parking Stand With Authenticated Locking System
- Done By: ALDRIN G (URK22EC1019)
- Contact : LinkedIn — Whatsapp — Email

Internship Project 2023

Problem Statement

Lack of security for cycles parked in the cycle stand in the hostels and public places

Solution

Our Solution Include

- A person has to scan his RFID tag in the RFID reader present in the bicycle stand, and an available slot gets allocated and when the person keeps his cycle in the allocated slot in the bicycle stand, it automatically gets locked after a second by a switch which gets triggered by the wheel of the bicycle when the person keeps the bicycle in the slot. To unlock the lock again he has to scan his RFID tag in the RFID reader and the lock of the allocated slot of the bicycle gets released and automatically closes after a second after the cycle is taken from the stand.

Tech Stack(Experience/Resources)

- RFID Reader and RFID Tags
- Bicycle Stand
- Servo Hook
- Arduino

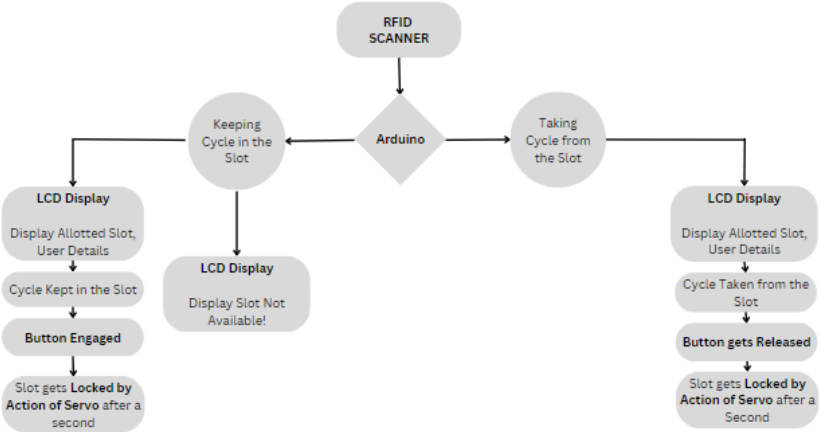
Summer Project 2023

Target Audience

- College Students and Students in Cities

Summer Project 2023

Work Flow



Future Plan

- Making it an IoT project using NodeMcu replacing Arduino
- Making a dedicated App, to monitor and control the system
- Using a Mechanical Lock instead of the Servo and Hook Mechanism